

ROANOKE-BLACKSBURG, VA, USA

WMO: 724110

Lat: 37.317N

Lon: 79.974W

Elev: 358

StdP: 97.10

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-6.8	-19.9	0.7	-5.4	-17.3	0.9	-3.9	12.3	2.9	11.2	1.6	4.1	310	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.4	22.6	32.1	22.1	30.8	21.8	24.1	30.2	23.5	29.3	23.0	28.3	3.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.8	26.3	21.8	17.2	25.8	21.3	16.7	25.5	74.1	30.5	71.7	29.5	69.5	28.5	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.7	7.6	-12.6	35.5	3.2	1.7	-15.0	36.7	-16.8	37.7	-18.7	38.6	-21.0	39.9		
			-13.8	25.3	3.0	0.6	-16.0	25.7	-17.7	26.0	-19.4	26.3	-21.5	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	2.9	4.7	8.9	14.4	18.8	23.1	25.0	24.3	20.7	14.9	8.9	4.9
	DBStd	8.90	5.74	5.33	5.41	4.71	4.08	2.98	2.55	2.55	3.45	4.33	4.79	4.81
	HDD10.0	748	231	161	87	13	1	0	0	0	0	8	76	172
	HDD18.3	2182	479	382	296	134	46	4	1	0	17	123	283	418
	CDD10.0	2332	11	13	52	145	274	392	464	444	320	161	44	12
	CDD18.3	722	0	0	3	17	61	146	206	186	87	17	1	0
	CDH23.3	5972	1	2	24	182	503	1173	1788	1529	634	126	9	1
	CDH26.7	2078	0	0	2	39	141	402	706	568	194	26	0	0
Wind	WSAvg	2.9	3.5	3.5	3.7	3.5	2.9	2.6	2.4	2.2	2.2	2.5	2.9	3.1
Precipitation	PrecAvg	1057	73	76	90	83	104	98	103	96	95	85	78	77
	PrecMax	1585	202	203	201	288	257	262	324	242	298	251	314	209
	PrecMin	634	7	14	9	12	26	16	11	19	4	0	5	5
	PrecStd	205	45	43	42	50	54	60	53	48	71	57	52	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	21.8	26.2	30.2	31.9	34.1	35.6	35.1	32.8	29.3	24.8	20.8
		MCWB	14.1	14.5	15.7	17.3	20.5	22.7	23.0	22.7	21.4	19.8	16.2	14.0
	2%	DB	17.2	18.6	23.3	27.7	29.9	32.1	33.5	33.0	31.0	26.6	21.4	17.4
		MCWB	13.1	11.7	14.0	16.5	20.1	22.2	22.9	22.6	20.9	18.5	14.2	12.8
	5%	DB	14.6	15.8	20.6	25.4	28.3	30.6	32.0	31.4	28.9	24.3	19.1	15.0
		MCWB	10.1	10.0	12.5	15.6	19.5	21.7	22.6	22.2	20.3	17.4	13.6	11.0
	10%	DB	11.7	13.4	18.1	23.1	26.3	29.1	30.6	29.9	27.1	22.2	16.9	12.6
		MCWB	7.6	8.2	11.3	14.8	18.6	21.0	22.2	21.7	19.7	16.4	12.0	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.0	17.0	19.9	22.6	24.1	25.1	24.8	23.4	22.4	18.3	16.6
		MCDB	18.5	20.4	23.9	25.4	28.7	30.5	31.7	31.3	28.8	26.2	21.3	19.7
	2%	WB	13.7	13.2	15.5	18.2	21.5	23.4	24.2	24.0	22.5	20.5	16.5	14.0
		MCDB	16.8	16.7	21.1	24.5	27.4	29.5	30.4	29.9	27.4	24.0	19.9	16.1
	5%	WB	10.9	11.1	14.0	17.0	20.6	22.7	23.7	23.4	21.8	19.0	14.8	11.7
		MCDB	13.5	14.5	18.9	22.8	26.1	28.6	29.6	28.8	26.2	22.9	17.9	14.2
	10%	WB	7.8	8.6	12.4	15.8	19.7	22.0	23.0	22.8	21.0	17.4	12.9	9.0
		MCDB	11.2	12.8	16.9	21.3	24.6	27.4	28.5	27.8	25.1	21.1	16.2	11.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.8	10.8	11.7	11.0	10.4	10.1	10.1	10.3	10.8	10.6	8.8
		MCDBR	12.3	14.1	14.5	15.0	13.3	12.2	11.9	12.1	13.0	13.4	14.0	12.3
		MCWBR	8.6	9.3	8.1	6.9	5.8	4.8	4.0	4.1	5.0	6.6	8.5	8.6
	5% WB	MCDBR	11.1	12.7	12.2	12.5	11.5	10.7	10.3	10.3	10.1	10.3	11.3	10.8
		MCWBR	9.0	9.5	7.8	6.7	5.5	4.6	4.1	3.9	4.5	5.9	8.4	8.9
Clear-Sky Solar Irradiance	taub	0.304	0.321	0.350	0.396	0.436	0.477	0.507	0.506	0.400	0.360	0.322	0.305	
	taud	2.494	2.432	2.401	2.293	2.192	2.116	2.057	2.032	2.329	2.433	2.503	2.520	
	Ebn at Noon	888	913	914	885	851	813	785	774	849	856	859	862	
	Edh at Noon	80	96	110	129	146	157	166	166	117	96	79	74	
All-Sky Solar Radiation	RadAvg	2.27	2.99	4.04	5.07	5.61	6.11	5.75	5.24	4.37	3.49	2.59	1.97	
	RadStd	0.18	0.36	0.26	0.40	0.49	0.43	0.37	0.26	0.43	0.42	0.23	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+91
		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+76

Nomenclature: See separate page